



SkyInsight

AI-POWERED AIRLINE INTELLIGENCE PLATFORM

A full-stack analytics and machine learning platform that transforms raw airline data into **actionable intelligence** — spanning customer experience, operational safety, sustainability, and strategic decision-making.

8

AI MODULES

5

ML MODELS

3

CNN CLASSIFIERS

129K+

RECORDS ANALYSED

Real-Time

ONNX INFERENCE

3-Tier

ARCHITECTURE

CORE MODULES



Decision Insights

Interactive BI dashboards surfacing KPIs, route performance, and loyalty segmentation directly from the airline data warehouse.



Satisfaction Prediction

Logistic regression ONNX model predicts passenger satisfaction probability from 22 service and flight features in real time.



Delay Prediction

ANN-based classifier forecasts flight delay risk using route, aircraft type, and scheduling signals with confidence scoring.



Loyal Circle

K-Means customer clustering by loyalty tier and churn risk. Radar charts and profile cards reveal four distinct traveller archetypes.



CO₂ Forecasting

Time-series regression model predicts carbon emissions per route, enabling airlines to optimise for sustainability targets.



Smart Vision

EfficientNetV2-S CNN classifiers detect luggage damage, cabin cleanliness issues, and crowd density from uploaded photos.



Reviews NLP

Web scraping → VADER & RoBERTa sentiment analysis → topic modelling → interactive visualisation in a single pipeline.



Performance Hub

Consolidated model metrics, business impact scores, and cross-module comparisons in one analytical overview.

PROJECT THEMES



End-to-End AI Ownership

From raw CSV ingestion to production ONNX inference — every layer designed and owned within the project.



Real-Time Intelligence

Sub-second predictions via FastAPI + ONNX Runtime, eliminating ML framework overhead at serve time.



Sustainability

CO₂ footprint modelling and route efficiency analysis supporting the industry's green aviation agenda.



Customer Centricity

Loyalty, churn, satisfaction, and review sentiment — every module keeps the passenger experience at its core.



Operational Safety

Delay forecasting and vision-based quality checks reduce operational risk and ground-level disruptions.



Data-Driven Decisions

BI dashboards connect data warehouse insights directly to frontline operators and strategic planners.

UN SUSTAINABLE DEVELOPMENT GOALS

SDG 9

Industry, Innovation & Infrastructure

- Smart route optimisation to maximise load factor
- CNN crowd analytics to reduce gate congestion
- ANN delay forecasting for schedule reliability
- Real-time operational KPI dashboard

SDG 12

Responsible Consumption & Production

- Fuel usage benchmarked per passenger-km per route
- CV cleanliness model reduces chemical cleaning waste
- Baggage QC vision cuts material damage and waste
- NLP pipeline for transparent service quality reporting

SDG 13

Climate Action

- CO₂ regression model per route, aircraft & season
- Year-on-year emission trend BI dashboard
- High-emission route flagging & offset engine
- Executive sustainability scorecard (Power BI)

TECHNOLOGY STACK

Python 3.11

FastAPI

ONNX Runtime

TensorFlow 2.21

scikit-learn

Next.js 15

TypeScript

Tailwind CSS

Recharts

HuggingFace

VADER NLP

SQL Server DW

Power BI

Git LFS

SkyInsight

AI FOR AIRLINES · BUILT WITH PRECISION

github.com/Larousse2001/SkyInsight

FULL-STACK · MULTI-MODEL · PRODUCTION-READY